Higher National Diploma in Engineering (Electrical and Electronic)
Fourth Year, First Semester Examination – 2020
EE 3215 – Broadband Communication

Instructions for Candidates:
Answer **any Five (5)** questions.
All question carry **equal** marks.

No. of questions : 06
No. of pages : 04
Time : **Three (03) hours**

1.

a) Define the terms of broadband communication? What are the advantages and disadvantages of the broadband communication? **(06 Marks)**

b) What is the difference between ISDN & ADSL broadband? **(03 Marks)**

c) What are the advantages of DSL over the cable modem? **(05 Marks)**

d) Briefly describe following terms

i. HDSL

ii. RADSL

iii. SDSL **(02 x 3 = 06 Marks)**

[Total 20 Marks]

2.

a) Draw the channel configuration for the ADSL? And briefly discuss that which types of multiplexing techniques is used in ADSL? **(04 Marks)**

b) Draw the connectivity diagram of ADSL connection showing different blocks from the central office to subscriber premises and explain the function of each block in the diagram. **(05 Marks)**

c) Fill in the table according to the ADSL data rates given below. **(06 Marks)**

Data Rate	Wire gauge	Wire size	Distance
1.5 or 2 Mbps			
1.5 or 2 Mbps			
6.1 Mbps			
1.5 or 2 Mbps			

d) What are the two types of modulation techniques used in xDSL technologies? Give a brief description for each illustrating the spectrum of each modulated signal? **(05 Marks)**

[Total 20 Marks]

3.

a) What is the need of ATM networks? **(04 Marks)**

b) Briefly describe what are the elements of the Asynchronous Transfer Mode (ATM)? **(04 Marks)**

c) What arte the applications of ATM and briefly explain them. **(04 Marks)**

d) Why ADSL is more popular than SDSL? Give both technical and marketing reasons? **(04 Marks)**

e) What is BPL and why it is not popular like DSL in many countries? **(04 Marks)**

[Total 20 Marks]

4.

a) What is XTALK in UTP cables used in DSL? **(04 Marks)**

b) How can the effect of XTALK be minimized in DSL communication? **(04 Marks)**

c) What is a loading coil and how does it affect DSL signals? Explain. **(04 Marks)**

d) Give three constraints that prevents the effective transmission of DSL signals on Copper wires and explain how the negative impact of each constraint can be minimized? **(04 Marks)**

e) If a DSL connection which was performing well initially has got poor performance after one year later, what improvements can be done on the external plant to get the good performance back? **(04 Marks)**

[Total 20 Marks]

5. Explain below situations.

a) Can ADSL connection work even without dial tone on the telephone? **(5 marks)**

b) Can the ADSL connection be used without the splitter/micro filter? What are the issues if you do so? **(5 marks)**

c) It is better to use a splitter at the entrance point of the subscriber line to the house and have a separate line to the ADSL modem, bypassing the house telephone wiring system. **(5 marks)**

d) There are number of factors that may affect broadband speed. Explain in briefly those factors. **(5 marks)**

[Total 20 Marks]

6.

a) Draw wireless broadband access network diagram using WiMAX as the access technology. **(04 Marks)**

b) What is base station in WiMax system? **(04 Marks)**

c) What are the role of internet service providers? **(04 Marks)**

d) What is the difference between the internet and the World Wide Web? **(04 Marks)**

e) Define various types of Internet connections? **(04 Marks)**

[Total 20 Marks]